

Equivalence Partitioning Test Cases for MembershipManager

Table 5

Variable	Valid/Invalid	Representative	TC01	TC02	TC03	TC04	TC05	TC06	TC07	TC08
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Test Condition 1 or Operation 1: calculateDiscount(int age, String membershipType)

Age Variable

Variable	Valid/Invalid	Representative	TC01	TC02	TC03	TC04	TC05	TC06	TC07	TC08
age < 18 (Children)	Valid	10	✓							
age >= 18 && age < 65 (Adults)	Valid	30		✓	✓	✓				
age >= 65 (Seniors)	Valid	70					✓			
age < 0 (Invalid)	Invalid	-5						✓		
age > 150 (Invalid)	Invalid	200							✓	

MembershipType Variable

Variable	Valid/Invalid	Representative	TC01	TC02	TC03	TC04	TC05	TC06	TC07	TC08
Premium	Valid	"Premium"		✓						
Standard	Valid	"Standard"			✓					
Basic/Other	Valid	"Basic"				✓				
Invalid Type	Invalid	"Gold"								✓
Any (for children)	Valid	"Standard"	✓							
Any (for seniors)	Valid	"Premium"					✓			

Test Condition 2 or Operation 2: isValidMembershipType(String membershipType)

MembershipType Variable

Variable	Valid/Invalid	Representative	TC01	TC02	TC03	TC04	TC05
Premium	Valid	"Premium"	✓				
Standard	Valid	"Standard"		✓			
Basic	Valid	"Basic"			✓		
Invalid Type	Invalid	"Gold"				✓	
Null/Empty	Invalid	null					✓

Expected Output

For calculateDiscount method:

Test Case	Age	MembershipType	Expected Discount	Rationale
TC01	10	"Standard"	0.2	Children get 20% discount regardless of membership
TC02	30	"Premium"	0.3	Adult premium members get 30% discount
TC03	30	"Standard"	0.15	Adult standard members get 15% discount
TC04	30	"Basic"	0.1	Adult basic/other members get 10% discount
TC05	70	"Premium"	0.4	Seniors get 40% discount regardless of membership
TC06	-5	"Standard"	0.2	Invalid age (negative) - treated as children
TC07	200	"Premium"	0.4	Invalid age (too high) - treated as seniors
TC08	30	"Gold"	0.1	Invalid membership type - treated as "other"

For isValidMembershipType method:

Test Case	MembershipType	Expected Result	Rationale
TC01	"Premium"	true	Valid membership type
TC02	"Standard"	true	Valid membership type
TC03	"Basic"	true	Valid membership type
TC04	"Gold"	false	Invalid membership type
TC05	null	false	Null input should return false

Summary of Equivalence Partitions

Age Partitions:

1. Valid Partitions:

- Children: age < 18
- Adults: $18 \leq \text{age} < 65$

- Seniors: age ≥ 65

2. Invalid Partitions:

- Negative age: age < 0
- Extremely high age: age > 150

MembershipType Partitions:

1. Valid Partitions:

- Premium
- Standard
- Basic

2. Invalid Partitions:

- Unknown membership types
- Null or empty strings

Key Testing Considerations:

- Case-insensitive matching for membership types
- Boundary values for age groups (17, 18, 64, 65)
- Combination of valid and invalid inputs
- Edge cases like null inputs